vii. Betwa, Chambal

viii. Ganga, Satluj

ix. Shipki-La, Nathu-La

x. Shimla, Kullu

xi. Chandigarh

xii. Arabian Sea, Indian Ocean

xiii. Tropic of Cancer

xiv. Shimla, Darjeeling

xv. Rajasthan, Madhya Pradesh

6

Our Community and Public Administration



We will learn in this lesson

- The meaning of democracy.
- Right to vote in elections.
- Governance in democracy.
- Rural and urban public administration.

Governance has been divided into three parts in the constitution. The Legislature, The Executive and The Judiciary. The Legislature means Parliament. It makes laws for the country. The Executive runs the administration of the country. The judiciary protects the constitution and laws.

Just like the Parliament in Centre, there are legislatures in each state. Apart from this there is a system of Panchayats at three levels. The three panchayats are - village level Panchayat, block level Panchayat and district level Panchayat. Members are elected by the people in the Lok sabha at the Centre, assemblies in the states and the three levels of panchayats. Panchayats perform development work at their own levels.

Just like in villages, there is a system of development in city areas also. In small towns and cities there are City councils. In bigger cities there are Municipal councils and there are Municipal corporations for the biggest cities. People elect members of City councils, Municipal Councils and Municipal corporations. Let us read more about these.

6.1 Democracy and Governance in India

India gained independence from the British on 15th August 1947. Before independence our country was divided into many small states. After independence all these states were unified and organized. Democracy was established in our country. The meaning of Democracy is a Government of the people, by the people, and for the people. The history of our modern democracy is not very old.

We have made many rules and laws to run the government. This is called The Constitution. This constitution became applicable on 26th January 1950. Every year 26th January is celebrated as Republic Day. 15th August is celebrated as Independence Day.

In a democracy the Government is run on the basis of majority. As such the entire structure of the country rests on the franchise of the people.

It is an unfortunate situation where on one side people demand greater rights for themselves and on the other side they do not use the rights given to them. In a democracy it is expected that citizens should vote on their own without any temptation or pressure. Until now many people have remained unenthusiastic about this right and hence in our country the voting average remains very low. More surprising is that many educated people also do not exercise their franchise. In a democracy every vote is very significant. Indian Voter Day is celebrated on 1st January. To be able to vote it is essential for every voter to enroll in the voters list.

There are 29 states and 7 Union Territories in India. The Central Government is above all the states and Union Territories. The states in India do not have the right to frame a separate constitution. Therefore our citizenship is single. We are citizens of India and not of any particular state. Every Indian has the same citizen rights irrespective of the state in which he/she lives.

The work of the Central Government has been divided into three parts - making laws for the country, having those laws implemented and reviewing the laws. These are called The Legislature, The Executive and The Judiciary.

6.1.1 Legislature

Legislature i.e. Parliament. The Parliament makes laws for the country. Our Parliament has two houses- Lok Sabha and Rajya Sabha. Laws passed by both houses of Parliament are implemented after the permission of the President.

Lok Sabha

Members of the Lok Sabha are elected directly by the people of our country. In our country all citizens of age above 18 years whose names are registered in the voters list, have a right to vote. Election Commission conducts elections for the Lok Sabha under its supervision. The Lok Sabha can have a maximum of 545 members. The tenure of the Lok Sabha is for five years. There cannot

be a gap of more than six months between two sessions of the Lok Sabha.



Parliament

In the Lok Sabha, the elected leader of the political party which has a majority becomes the Prime Minister. The prime Minister makes his Government. If one party does not have a majority then it has to take support from other parties. In such a situation a coalition government is formed. The leader who gets the backing of more MPs becomes the Prime Minister. The elected leader of the party with a majority becomes the Prime Minister even if he is a member of Rajya Sabha or not of Lok Sabha. After becoming Prime Minister, within six months he must become a member of one of the Houses of Parliament. He forms a cabinet with members of coalition parties. The Council of Ministers of the coalition runs the administration of the country. The party that forms the Government is called the ruling party. Those parties whose MPs are not in Government are called opposition parties.

Rajya Sabha

Members of Rajya Sabha are elected by legislators of state assemblies. They are elected in an indirect manner. Members of Rajya Sabha are also called MPs. In the Rajya Sabha, the strength of members from each state is determined on the basis of the population of that state. There is a total of 250 MPs in the Rajya sabha. Out of these there are 12 nominated members. Rajya Sabha is the House which remains in continuous existence. After every two years, one third of the members retire. In this way every MP has a tenure of six years.

6.1.2 The Executive

President, Vice President, Prime Minister, Cabinet and Government employees together form The Executive. The entire administration of the country is run by The Executive. The Prime Minister is the Chief of The executive. Because India is a Republic, therefore the entire functioning of the government is done in the name of the President. All Government officers are employees who work under him. For the smooth working, all rules are made in the name of the President. But the President is only a Constitutional head. In practice all powers are exercised by the Prime Minister. The Prime Minister executes all powers of the president through his cabinet.

6.1.3 The Judiciary

The Judiciary is responsible for the protection of the Constitution and laws of the country. The Supreme Court is the highest court of the country. It has one Chief Justice. Judges are appointed by the President. The Judiciary interprets the laws. It protects the fundamental rights of the citizens also. Any ruling of the Supreme Court has to be followed by all the courts.



Supreme Court

6.1.4 Federal Governance

In our country there is a system of Federal Governance. In the Constitution there are rules for making Parliamentary forms of Governments at the Centre and States. The state administrative system supports the Government functioning at the Centre. In the states also people elect members to the state assemblies. Members to the Legislative Councils are elected in an indirect manner. Just like the Lok Sabha the tenure of the state assemblies is also for five years. Just like the Rajya Sabha, the tenure of the Legislative Councils is of six years. In many states there are no Legislative Councils.

The Constitution has provided different powers for the Centre and States. There is provision for President, Prime Minister and Parliament in the Centre and Governor, Chief Minister and Legislative assemblies in States. In the judicial area, High Courts have the same powers as the Supreme Court. In the States also just like in the Centre the administration is run within the state limits. Both the Centre and States have the powers to frame rules in their respective areas. In some matters both can frame their own laws. In case of a clash between the Centre and States, the laws of the Centre will prevail. In the states, Governors work as representatives of the President. In states, the Governor is the chief of The Executive. The Governor is appointed for a period of five years. The president can remove him from his post at any time. It is the duty of the Governor to ensure that there is a stable Government in the State.

Let's see, what you have learnt 6.1

I. Fill in the blanks:

i. In India there are 29 states and ..... Union Territories.

ii. India became independent on .....

iii. Our country is a Federal .....

iv. In the Lok Sabha the leader of the majority becomes the .....

v. The working of the Government has been ..... into three parts.

II. Match the Columns:

i.	Legislature	a.	Citizenship
ii.	Single	b.	Parliament
iii.	Rajya Sabha	c.	Judiciary
iv.	Supreme Court	d.	Lok Sabha
v.	5 years	e.	6 years

III. Choose the correct word and fill in the blanks:

Executive Council Democracy 5 years Laws

i. Centre and states have the powers to make .....

ii. In many states there is no Legislative .....

iii. The administration of the country is run by .....

iv. After independence ..... was established in the country.

v. The Governor is appointed for .....

6.2 Rural Public Administration

6.2.1 Rural Public Administration

The management of the administration of villages in India is done by the local people there. This is called local self governance. The meaning of self governance is that the local people run their own administration. Therefore in rural areas the administration is run by area, block and district Panchayat. Today more than two lakh Panchayats are working in India. All Bodies are elected by the people. They provide facilities to the people in accordance with the rules and regulations of the country. To do this they work in their own areas.

6.2.2 Village Panchayat

Village Panchayat is established in villages with a population of more than one thousand people. If the population is less, then nearby villages are also included in that Panchayat. Therefore there can

be one or more than one village in a village Panchayat.

Village Panchayat is constituted by three parts.

1. Gram Sabha
2. Village Panchayat Committee
3. Village Panchayat

Gram Sabha



Gram Sabha Meeting

This is the biggest gathering of the village. Every citizen of the village of over 18 years of age is a member of it. The residents of the village present their ideas on the problems and development work in the village to the village Head and members of the Panchayat committee. They can also ask questions about their work. There should be at least four meetings of the Gram Sabha in a year. They should be held on 26th January, 1st May, 15th August and 2nd October of every year. To complete the quorum of these meetings at least 10 percent of the total members of the Gram Sabha should be present. The meeting of the Gram Sabha is presided over by the Head or in his absence by the Deputy Head. The Gram Sabha passes its resolutions by a majority of the members present.

- Annual income and expenditure and accounting of the work done.
- Previous year's administrative Report.
- Audit Report
- Next year's Budget
- Development programmes.
- Programmes to be started in the current year.
- Surveillance committee's Report

Village Panchayat Committee

This is a committee of the elected representatives of the village. All residents of the village whose names are present in the voters list elect the village Panchayat members and the village Head Sarpanch. To be elected as a member, the name of the person should be in the village voter list and he/she should be at least 21 years of age. The number of members of the Panchayat committee is

according to the population of the village. This number can be a minimum of 9 and a maximum of 15. For a population of one thousand, 9 Panches, for a population of one thousand to two thousand, 11 Panches, for a population of two thousand to three thousand, 13 Panches and for a population of over three thousand, 15 Panches are elected.



Gram Sabha Meeting

This committee should have one meeting every month. It is presided over by the Village Head/Sarpanch. It is essential to have backward classes and women in this committee. One third of the positions are reserved for women. The number of Panches belonging to Scheduled Castes and Scheduled tribes is determined on the basis of their population in the village. The members of the Panchayat committee elect a Deputy Head from amongst themselves.

Village Head

The village Head is the Chief of the village. He is elected by all the residents of the village. He is also the ex officio member of the Area Panchayat. His tenure is for five years. In some states, there is a provision for payment of salary to the Village Head, Deputy Head and members of the committee for their services but it does not apply to all places. The Gram Sabha can appoint employees to help in its work. Their salary is paid by the Gram Sabha.

Qualification of Panches

1. Age should be at least 21 years.
2. Should be a voter of that Village Panchayat area.
3. Is not disqualified for election under any law.

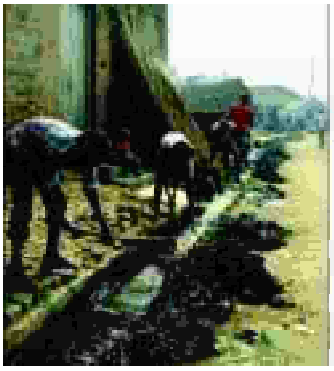
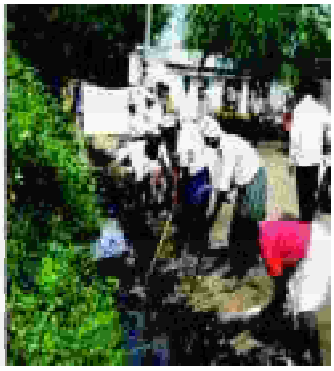
Tenure

The tenure of the village Panchayat is of five years. The Head can be removed before the end of his term by the Panches by passing a motion by a two thirds majority. This motion is called a no confidence motion. But for this motion, a meeting cannot be called within one year of the holding of election.

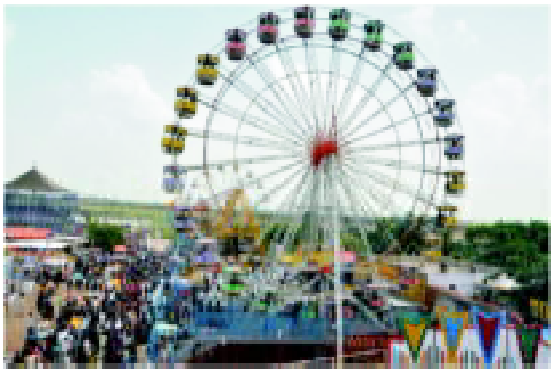
6.2.3 Duties and Rights of Village Panchayat

1. Making complete arrangement for light and cleanliness of all public places.
2. Making arrangements for pure and clean drinking water.

3. Registration of Births and Deaths.
4. Making available health services.
5. Making arrangements for pension etc. for widows, helpless and old persons.
6. Opening of fair price shops.
7. Making arrangements for fairs, festivals and other means of entertainment.

Village Cleaning



fairs



Tree Plantation

8. Increasing the usage of biogas and smokeless earthen stoves.
9. Stopping and removing illegal occupation of public property.
10. Development of forests and tree plantation.
11. Development of barren land and pastures for grazing.
12. Development of village and cottage industries.
13. Allotment of land for construction of new residential houses.

6.2.4 Area/Block Panchayat

Many Village Panchayats are combined to form an Area or District Panchayat. This is also called Block Panchayat. To elect members of Block Panchayat, the Block is divided into many parts. From each part, the local people elect one member. He is called Block Panchayat member. Some seats are reserved for backward classes, Scheduled Castes and Schedule tribes and women. Apart from this, all Village Heads, MPs and MLAs of that area are also members of Block Panchayat.

Block Panchayat members elect one Block Panchayat President (Block Head) and two Vice Presidents from amongst themselves. The tenure of the Block Panchayat committee is of five years. All development work in the Block is done under the supervision of the Block Panchayat.

6.2.5 District Panchayat

Every District has one District Panchayat. It does all development work of the District. To elect members of District Panchayat, the District is divided into different constituencies. From each constituency one member is elected to the District Panchayat. All Block Heads, area MPs and MLAs are members of the District Panchayat. Members elected by the people elect one Block President and one Vice President from amongst themselves. The tenure of the District Panchayat is of five years. In the absence of the President, the Vice President looks after his work. The District Panchayat makes plans for the development of the District and makes arrangement for financing the same.



Let's see, what you have learnt 6.2

I. Fill in the Blanks:

i. All citizens who are over 18 years of age are members of .....

ii. In the meeting of the Gram Sabha the quorum is ..... percent of the members.

iii. To become a member of the Village Panchayat committee the minimum age should be ..... years.

iv. Members of ..... elect a Deputy Head from amongst themselves.

v. For a no confidence motion, a .....cannot be called within one year.

II. State whether the following are Correct or Wrong and mark accordingly.

i. Self rule means the local people run the administration themselves. ( )

ii. It is essential to have one meeting of the Gram Sabha every month. ( )

iii. To be elected a Panch the minimum age should be 21 years. ( )

iv. Registration of Births and Deaths is not the work of Panchayat. ( )

v. Panchayat has a right to collect money for its work. ( )

6.3 Urban Public Administration

Just like in rural areas, there is a provision for self rule in urban areas also. For this there are City Councils, Municipal Councils and Municipal Corporations. They work in their notified areas. Voters elect members for them from their respective wards.

6.3.1 City Council

The City Council is constituted on the basis of the population of the city. The City Council consists of members elected by the people. To elect them the city is divided into Wards. Voters of every Ward elect their representative. The entire city elects the President of the City Council. His tenure is of five years. The City Council makes arrangements for cleaning of Roads, lanes and drains of its area. It has to make arrangements for lighting of roads and education etc.

6.3.2 Municipal Council

In cities where the minimum population is 3 Lakhs and the maximum population is 5 Lakhs, a Municipal Council is constituted. Residents of these cities send their elected representatives to the Municipal Council. The minimum age for becoming a member of the Municipal council is 25 years. There can be 15 to 60 members in a Municipal Council. Some seats are reserved for backward classes, Scheduled Castes and Scheduled Tribes and women. All voters of the city together elect the President of the Municipal Council. From time to time for the problems and development work of the city the President calls meetings of the members. In the absence of the President, the Vice President looks after his work.

6.3.3 Municipal Corporation

If the population of a city is above 5 Lakhs then a Municipal corporation is constituted. To elect members of the Municipal Corporation, the city is divided into different Wards. From every ward one member is elected. There can be 60 to 150 members in a Municipal Corporation. The elected members of the Municipal Corporation elect a president. In some states the President is elected directly by the people. The President is also called the Mayor. The members also elect a Deputy Mayor from amongst themselves. The Mayor presides over the meetings of the Municipal Corporation. To make the life of its citizen healthy and comfortable, Municipalities and Municipal Corporations in their respective areas try and do all types of work for the public in a smooth manner. Their tenure is also of five years.

With the basic purpose of making urban life full of facilities, these local bodies are constituted. To perform all these activities these local bodies have to collect money. To do this they have the powers to collect some taxes. Apart from this, the local bodies also get grant (money) from the State Governments.

Let's see, what you have learnt 6.3

I. Fill in the blanks:

i. In urban areas also there is a system of .....

ii. In Municipal Councils one third of the seats are .....for women.

iii. The entire city ..... the City Council President.

iv. Local Bodies get ..... money from State Governments.

v. The tenure of the Mayor is of ..... years.

II. Match the Columns:

i.	City Council Member	a.	Money
ii.	Ward	b.	Over meetings
iii.	Grant	c.	Clean
iv.	Drains	d.	5 years
v.	Presided	e.	Voter

III. Tick the correct word with a (✓) :

i. How many members are elected from each ward? One member/six members

ii. What is the age of an adult citizen? 10 years/18 years

iii. Who is the President of the Municipal Corporation? Commissioner/ Mayor

iv. What is the minimum age of a member of Municipal Council? 25 years/21 years

v. How is money collected for development work? From donation/ From taxes

Let's Revise

- In a democratic system the Government is of the people, by the people and for the people.
- The work of the Central Government is divided into three parts -

Legislature -- Makes laws for the country

Executive -- Implements those laws

Judiciary -- Reviews the constitution and Laws

- Our Parliament has two Houses - Lok Sabha and Rajya Sabha. There are 545 members of the Lok Sabha.
- In our country citizens of the age of 18 years and above have a right to vote.
- In the Lok Sabha the leader of the party which has a majority is elected as the Prime Minister.
- There are 250 members in the Raja Sabha. 12 members are nominated.
- The tenure of the Lok Sabha is of 5 years. The tenure of each member of the Rajya Sabha is 6 years.
- The President, Vice President, Prime Minister, Cabinet and Government employees together are called the Executive.
- The responsibility to protect the Constitution and laws of the country lies on The Judiciary.
- The Village Panchayats are constituted by combining three parts -
  1. Gram Sabha
  2. Village Panchayat members
  3. Village Head
- The tenure of Disatrict Panchayats is of 5 years.
- The minimum age to become a member of the Municipal council is 25 years.
- There can be 15 to 60 members in a Municipal Council.



Exercise

I. State whether the following statements are correct (✓) or Wrong (✗)

i. Just like Parliament there are Legislatures in States. ( )

ii. On 26th January 1950 India became independent. ( )

iii. Parliament makes laws for the country. ( )

iv. The Judiciary runs the administration. ( )

v. For big cities there are Municipal corporations. ( )

II. Match the Columns:

i.	15th August	Lok Sabha
ii.	Rajya Sabha	Independence

iii.	545 members	6 years
iv.	Governor	Panchayats
v.	2 Lakh	5 years

III. Fill in the Blanks:

i. The administration of the country is run on the ..... of majority.

ii. The work of the Central Government has been divided into ..... parts.

iii. The two Houses of Parliament are Lok Sabha and .....

iv. Village Head is the ..... of the village.

v. The management of the village is done by their .....

IV. What do the three arms of the administrative system do in the Constitution?

Legislature - _____

Executive - _____

Judiciary - _____

V. Tick the correct answer:

i. Voting right means.

Right to vote / Right to go abroad

ii. Age of becoming a Major

16 years/ 18 years/ 20 years/ 22 years

iii. The average of voting in India is

Normal/ low/ very high/ very low

iv. Franchise in democracy is

Unimportant/ important/ mandatory

VI. What is the meaning of self rule in a democracy?

_____

_____

_____

VII. What is the age of becoming a major in India? What was this age before?

_____

_____

_____

VIII. On which days is it mandatory for Village Panchayats to have their meetings?

_____

_____

_____

IX. Which issues can be discussed in a Gram Sabha? Mention three issues.

_____

_____

_____

X. Name the Bodies that run the administration in rural and urban areas.

_____

_____

_____

Come let's do

Go to the local bodies in your village or city and see how your representatives do their work.

ANSWERS

Let's see what you have learnt

6.1 I. Fill in the blanks:

i. Seven
ii. 15th August 1947
iii. Republic
iv. Prime Minister
v. Divided

II. Match the columns

i.	Legislature	a.	Parliament
ii.	Single	b.	Citizenship
iii.	Rajya Sabha	c.	6 years
iv.	Supreme Court	d.	Judiciary
v.	5 years	e.	Lok Sabha

III. Choose the right word and fill in the blank.

i.	Laws	ii.	Council
ii.	Executive	iv.	Democracy
iii.	5 Years		

6.2 I. Fill in the blanks

i.	Gram Sabha	ii.	Ten
iii.	21	iv.	Panchayat committee
v.	meeting		

II. State whether the following are Correct or Wrong and mark accordingly.

i.	Correct	ii.	Wrong
iii.	Correct	iv.	Wrong
v.	Correct		

6.3 I. Fill in the blanks

i.	Self rule	ii.	reserved
iii.	elects	iv.	Grant
v.	five		

II. Match the Columns

i.	City Council Member	a.	5 years
ii.	Ward	b.	Voter
iii.	Grant	c.	Money
iv.	Drains	d.	clean

v. Presided e. Over meetings

III. Tick the correct word with a (✓)

i. One member ii. 18 years

iii. Mayor iv. 25 years

v. From taxes

Exercise

I. State whether the following statements are correct (✓) or Wrong (✗)

i. Correct ii. Wrong

iii. Correct iv. Wrong

v. Correct

II. Match the Columns

i.	15th August	Independence
ii.	Rajya Sabha	6 years
iii.	545 members	Lok sabha
iv.	Governor	5 years
v.	2 Lakh	Panchayats

III. Fill in the Blanks

i. Basis ii. Three

iii. Rajya Sabha iv. Chief

v. Citizens

IV. Tick the correct answer

i. Right to vote ii. 18 years

iii. low iv. important

Assessment Sheet 2

From Lesson 4 to 6

7

Life of Plants and Animals



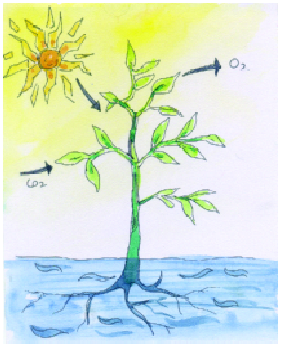
We will learn in this lesson

- How plants and animals are dependent on one - another?
- What is food chain and food trap?
- What is natural balance?
- Effects if the natural balance is disturbed.
- Why is environment protection necessary?
- Different methods of protecting forests.
- How animals can be protected?

Look around, wherever you see plants and trees, you will also find several types of animals. In nature, can you imagine any place where there are just trees and plants but no animals or where there are animals but no plants? Actually, in this animate world, everywhere there is vegetation and also animals, both are an inseparable part of this animate world.

Plants and animals are dependent on one - another. Together they keep a balance in nature. Many deeds, of man harm this balance. This affects the whole flora and fauna. Therefore, we will have to work on different measures to maintain a balance in nature. In this lesson we will learn about the relationship between plants and animals, its balance and protection.

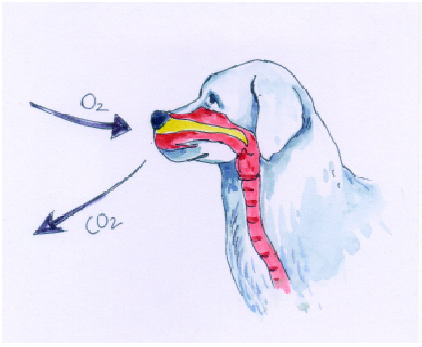
7.1 Plants and animals are dependent on each other



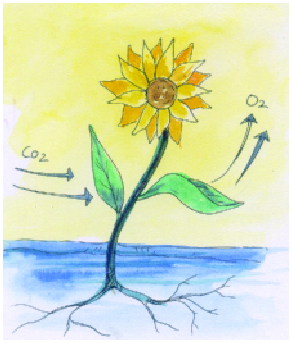
Plants and animals are dependent on each other for different things, for example, all animals obtain their food from vegetation.

In nature only the green plants have the ability to absorb air and water and obtain energy from the sun and turn it into food. All animals are dependent on plants for their food. Think, how?

Plants and animals together balance the amount of oxygen and carbon -dioxide in the air.





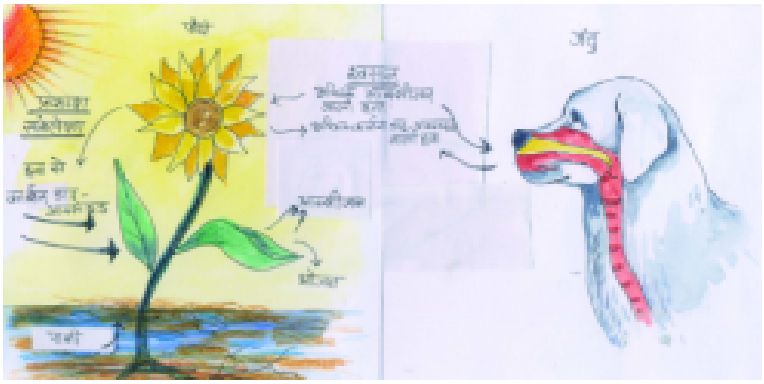


Oxygen in the air

Plants and animals through their breathing inhale oxygen and exhale carbon - dioxide. This action is continuous (day - night) but in the air, neither does the amount of oxygen become less and nor does carbon - dioxide increase too much. Why so?

This is because the carbon - dioxide exhaled through our breathing is absorbed and utilized by the plants. The green leaves of the plant, make food with the help of sunlight, water and carbon - dioxide.

This process of making food is called 'Photosynthesis'. It leads to the generation of oxygen gas. This way the amount of oxygen in the air does not decrease and maintains the balance between oxygen and carbon - dioxide in the air.



Photosynthesis

Trees and plants provide shelter to animals.

- A number birds, insects, big and small animals like to live on trees. In dense forests the wild animals get a place to live and hide themselves among trees and plants.
- A number of insects and birds help in pollination. To transfer the pollen of one flower to another is called the process of pollination.

In fact, insects and birds sit on the flowers to drink its nectar. The pollen from the flowers sticks to their bodies (legs, wings and face). When these insects and birds fly and sit on other flowers the pollen sticking to them drifts down on the other flowers.

Birds and other fauna take the seeds of the flowers far and wide.

- Birds and animals eat the fruits of the plant along with its seed. Wherever they pass their excreta, the seeds of the plant come out in the excreta and grow.
- In the seeds of some plants, there are thorns or small-small knots. These seeds stick to the skin of animals and fall at different places and there they grow. This way the seeds of the plants grow at far flung places.



Birds and other fauna take the seeds of the flowers far and wide.

Let's see, what you have learnt 7.1

I. Fill in the blanks:

i. Leave make food from ..... and ..... and sunlight.

ii. Plants and animals in the act of breathing, inhale ..... and exhale .....

iii. The process of taking pollen from one flower to another is called .....

7.2 Food chain and food - trap

Food is every creature's necessity. As we know, only green plants are able to make their food with carbon -dioxide, water and sunlight .Rest of the creatures, in some way or other are dependent for their food on these plants. Let's see, how in nature, one creature gets food from other creatures.

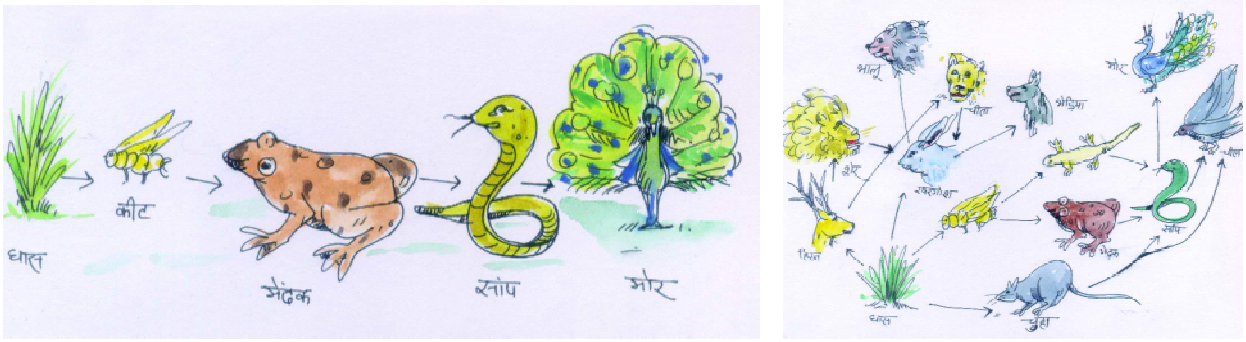


Food Chain

In this photo you are seeing a deer eating grass and the lion eating the deer. We can write this chain as:

Grass ⟶ Deer ⟶ Lion

This way a chain related to food is called a food chain. In this photo many more food chains are given. Let's see who is whose food. On earth these food chains are found in all places (fields, mountains, rivers, oceans). In our environment many food chains are inter - linked and form a net like shape. This is called a food - trap.



Food Trap

You have seen that the beginning of every food chain is vegetation. Creatures who eat vegetation are called herbivores. Creatures who eat the meat of other creatures are called carnivores. Cows, buffaloes, deer etc are herbivorous animals and lion, tiger, hyena etc are carnivorous animals. Those creatures which eat both vegetation and animals are called omnivores, like man.

Those creatures which feed on the dead body of other creatures are called scavengers like kite, vulture, crow, hyena, jackal etc.

They eat the meat of the dead animals, which cleans the environment.

Let's see, what you have learnt 7.2

I. Fill in the blanks:

i. In nature all food chain start from .....

ii. Many food chains link together and make a .....

iii. Those creature which eat the meat of other creatures are called .....

iv. Those who eat the meat of dead creatures are called .....

II. Write, which creates come under which group.

Kite, horse, buffalo, tiger, elephant, mouse, dog, deer, fox, leopard, frog, man, cat, jackal, vulture, lion.

Herbivores	Carnivores	Omnivores	Scavengers
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....

7.3 Natural Balance

In nature, creatures eat each other. Vegetation is eaten by herbivores, herbivores are eaten by carnivores, carnivores and herbivores are eaten by scavengers.



Scavengers eating dead bodies

The residues of plants and remains from the flesh of creature etc. are mixed in the soil by bacteria. Bacteria are very tiny germs which cannot be seen by our naked eyes. To look at them we use a microscope.

7.4 Consequences of Imbalance in nature

In nature, carnivore creatures prey on the herbivores and keep their numbers in control. Just think, if the number of carnivores decreases, then what will happen?

- The number of vegetarians will increase.
- To feed the vegetarians more trees and plants will be needed. This will reduce the number of trees and plants.
- Vegetarians will dig the roots of the plants and eat them and if they are unable to find any vegetation then they will start dying of hunger.
- This will harm all the creatures.

As it is, animals, are eaten in greater number which the reproduction capacity of these animals is more. This means they give birth to a large number of children. A rabbit gives birth to 6 - 8 babies at a time, while a lion gives birth to 1 or 2 cubs only.



Carnivores Family



Herbivores Family

You must have understood by now what problems can rise if we hunt wild animals in greater number.

Just think if we keep cutting our forests then how it will affect the balance in nature.

- Where will be the wild animals habitat and what will they feed on?
- In the forest, water from the leaves of plants becomes vapour and mixes with air. This forms clouds and we get rainfall. If forests are cut, then how will it rain?

- In the forests, the rain water hits the trees and moves forward at a show pace. This slow moving water turns into small waterfalls and flows from springs. If the forests are cut then the rain water will flow en masse and there will be danger of floods. If it does not rain, there will be drought.
- We get oxygen from trees and the air remains clean. The cutting of trees will increase the carbon - dioxide in the atmosphere and pollution will increase.

The environmental imbalance will endanger the life of man and animals. To prevent this, protection of environment is necessary.



Let's see, what you have learnt 7.3

Fill in the blanks:

i. If there are less carnivore animals then the number of ..... will increase.

ii. ..... are very small creatures which our naked eyes cannot see.

iii. In the forest the rain water hits the ..... and moves forward .....

iv. Animals which are an easy prey, their ..... is also more.

7.5 Measures to protect the forests

You must have heard about the Bishnoi caste of Rajasthan. It is said, once a king ordered the cutting of all trees of their area. The Bishnoi community clung to the trees and said, first you will have to cut us and only then the trees. The king's soldiers, following his orders, actually killed some people. When the king heard this he felt very bad. He never again thought of cutting trees in the areas inhabited by the Bishnois. Just think, why did the Bishnois not allow the trees to be cut? Should we cut trees thoughtlessly?

Plant as many trees as you can

These days the government has put a stop to cutting of trees. If we want to cut trees in the forest, then we have to take permission. We have to plant saplings in place of trees which have been cut.

How can animals be protected?

- By putting a stop to hunting of animals.
- By do not using articles made from animal skin.
- By making special shelter homes. National parks and exercise areas for animals.

In our country there are many forests where wild animals are specially protected. They are called national parks wild life sanctuaries.

Let's see, what you have learnt 7.4

I. Write two ways to protect forests?

_____

II. What are Bishnoi's known for?

_____

III. What are national parks or wild life sanctuaries?

_____

Let's Revise

- Plants and animals are an integral part of the living world.
- Plants and animals depend on each other for different activities.
- All animals in some form or the other get their food from plants.
- Vegetation and creatures together to maintain a balance of oxygen and carbon - dioxide in the air.
- Vegetation is the habitat of many animals.
- Many kinds of insects, birds etc help in pollination of flowers or plants.
- Creatures help to spread the seed of plants far and wide.
- It is necessary to protect plants and trees.

Exercise

I. Fill in the blanks in the following food chain.

i. Plant → Rabbit → .....

ii. ..... Mouse → ..... → .....

iii. Plants → Insects → Frog → ..... → Kite

II. Write two example of each

i. Herbivorous creatures _____ , _____

ii. Carnivorous creatures _____ , _____

iii. Omnivorous creatures _____ , _____

iv. Scavengers _____ , _____

III. Write two bad-effects of cutting of trees.

_____

_____

_____

IV. How can we protect animals? Give three measures.

_____

_____

_____

V. i. What is Photo-synthesis?

_____

_____

_____

_____

ii. What is the process of pollination in plants?

_____

_____

_____

_____

ANSWERS

Let's see what we have learnt

7.1 i. Water, Carbon - dioxide

ii. Oxygen, Carbon - dioxide

iii. Pollination

7.2. I. i. Plant

ii. Food Trap

iii. Carnivores

iv. Scavengers

II.

Herbivores	Carnivores	Omnivores	Scavengers
Horse	Tiger	Mouse	Eagle
Buffalo	Leopard	Dog	Hyena
Elephant	Frog	Human Beings	jackal
Deer	Lion	Cat	Vulture

7.3 i. Vegetarians

ii. Bacteria

iii. Plants, Slowly - Slowly

iv. Numbers

7.4 I.
- By preventing cutting of trees.
- By planting more trees.

II. Bishnoi community is known for protecting trees and plants.

III. National parks and wildlife sanctuary are those forests. Where wild animals are especially protected.

Answers

I. i. Lion

ii. Grains (Plants) Cat, Dog

iii. Insects, snake

II. i. Cow, Buffalo

ii. Tiger, leopard

iii. Human Being, Dog

iv. Kite, Hyena

III. • Less Rain Fall

• Increase in danger of floods and drought.

IV. • Stop hunting of animals.

• Do not use articles made from animal skin.

• Build National parks and wildlife sanctuaries.

V. i. The green leaves of the plant, with the help of sunlight, water and carbon - dioxide make food. This process of making food is called 'Photosynthesis'

ii. Insects and birds sit on the flowers to drink its nectar. The pollen from the flowers sticks to their legs, wings and face. When these insects and birds fly and sit on other flowers the pollen sticking to them drifts down on the other flowers. To transfer the pollen of one flower to another is called the process of pollination.

Come, let's see

To make the people aware of preservation of animal and forests, write two slogans.

8

Soil and Crops



We will learn in this lesson

- Types of soil
- Which soil is good for different crops?
- How to prepare the farm before sowing?
- After sowing how to manage your field.
- Things to be kept in mind before giving fertilizer and manure.
- Provisions to be made to protect your crop from what all things.
- Importance of irrigation, weeding, hoeing, harvesting and mounting.
- Which crops can be sown in different seasons?

Soil or in other words earth. Our body is said to be created from mixing with five elements. One of these five elements is earth. Earth is the basis of life. All kinds of cereals, fruits, flowers and medicinal herbs are grown in the soil. Their consumption is keeping us alive and healthy. Soil and crops have an unbreakable bond. The harvest depends on the soil. How can we get a good produce from harvest? What kind of crop can be grown in what type of soil? What is necessary for crops? Which season is suitable for which crop? We must have all this information if we want to do good farming.

8.1 Soil is Priceless

We know very well how priceless soil is for us. All kinds of cereals, vegetables, fruits and flowers are grown in the soil. Just think what would be there if there was no soil?

Come, let's undertake an experiment

Bring a little soil from the field. Fill water in a glass tumbler. Mix the soil in it with a spoon and leave it for sometime. We will see some things floating on the top surface of water. These are small pieces of straw, pieces of leaves, fragments of animals and rotten substances. Those things which are floating are called humus. This humus makes the soil soft and fertile.



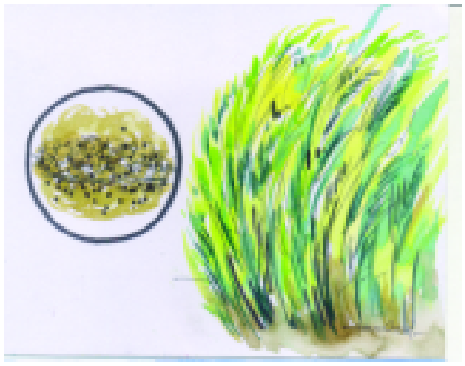
Checking the soil

Humus is found in the top layer of the soil. The top soil is the most fertile. That is why the top layer of the soil is called a farmer's wealth.

Look closely at the water in the glass. The bigger particles of soil have settled down while the smaller ones are still floating in water.

The soil is not the same in all places. Different places have different soil. According to the land formation our country has four main types of soil.

a) **Clay soil:** In this kind of soil the particles are very small. Between the particles there is very little empty space. It has the capacity to hold water for a longer period. It is also called fuller's earth. In this kind of soil, cotton, rice, sugarcane and gram have a good produce.



Clay soil

b) **Sand soil:** In this soil there are 60 percent or more particles of sand. This soil has the capacity to soak a lot of water but the capacity to retain it is less. There is a lot of space between particles. The particles lack the capability to stick to each other. For this reason the soil is very powdery. It is easy to do weeding, hoeing and cultivating on it. Sandy soil is also called light soil. In this soil, water melon, melon, potatoes, sweet potatoes, groundnut type of crops give a good produce.



Sand Soil

c) **Alluvial Soil:** Alluvial soil is one between sandy soil and clay soil. It has the quality of both the light and heavy soil. Therefore, It is suitable for most of the crops.

d) **Acidic soil:** In many places due to heavy rainfall, minerals such as calcium and magnesium filter down along with water to the depth of the soil. This makes the top soil very acidic. It is also called infertile land. This type of soil is not suitable for farming. In this kind of soil, we can mix lime and gypsum to make it fit for farming.



Acidic Soil

Let's see, what you have learnt 8.1

I. What is called the wealth of the farmer?

_____

II. Which soil has the maximum capacity to hold water from flowing?

_____

III. Which soil gives us the best produce of groundnuts?

_____

IV. How can we make acidic soil suitable for farming?

_____

8.2 Good Crop Abundant Harvest

- We eat food, drink water, take care of our body, bathe and wash, clean it and take medicine if sick. Why? So that we keep healthy.
- In the same way crops also have some requirements. They also require nutrients to keep healthy such as water, air, sun and proper care.

Any lapse or deficiency will make the crop weak and if the crop is weak than the harvest will be deficient. For a good harvest it is important to know and understand the needs of a good crop.

8.2.1 Sowing of Crop

Sowing must always be on time. If we sow before time or after its time then the seeds do not germinate properly. The growth of roots and sprouting will be less. This may lead to an outbreak of disease or attack on crops by insects. This will decrease the harvest.

8.2.2 Preparing the Farm

Before we sow the crop, we should prepare our farm properly. In making the preparation, the main work is of cultivation, hoeing and tiling the land. Cultivating the land makes the soil powdery and the air travels properly. This increases the capacity to soak water and we are also rid of harmful weeds and insects.



Preparing a farm

We must remember one thing about cultivation. When there is proper moisture or oat (a particular condition) only then cultivate the farm. If we till the land when there is more moisture the soil will not be granular, mounds will form in the soil and the air will not move freely. Germination and sprouting of seeds will be less. Field is raw before the arrival of oat.

There is a saying by Ghagh.

"Kacha Khet Na Jote Koi"
"Nahi Beej, Na Ankur Hoi"

(Means - Do not till a raw farm. It will hold neither seeds nor sprouts.)

- Therefore before cultivating we must check if there is proper moisture or oat. This maturates the field. To check, we must dig with a hoe or trowel and take out two or three small lumps.

Squash them with your hands, if the lump crumbles and falls then presume there is moisture in it. The farm is ready to be tilled. If the lumps do not break properly, or the particles in the soil do not separate then the farm is still raw.

- Remember, when the field is moist we must cultivate it on time. If the soil dries up more then we will not be able to do proper tilling. When we are preparing the field we should mix manure and compost in the soil according to the requirement of the crop. If necessary, then before sowing, we should treat the soil with proper pesticides. The treatment of the land removes many afflictions of the soil.

8.2.3 Seed Treatment

Before sowing it is beneficial to refine and treat the seeds. There are a number of chemical drugs like this Carbondazim, Thirum, Captain etc. for treating seeds. The treatment will prevent some seed borne or soil borne diseases. Apart from this, there are many bio-fertilizers such as Rhizobium Culture, P.S.B. culture and Tricoderma etc.

Seed treatment makes the seeds grow uniformly and healthy. There is increased germination. Nutrients are saved and the yield is higher.



Seed Treatment

8.2.4 Weeding and Digging



Weeding and digging

Usually, the weeds also grow in the fields when crops grow. If we do not remove these weeds then they will absorb all the manure or water meant for the crops. In comparison to crops, weeds grow faster and they can flatten the crop and reduce the harvest. Timely weeding and digging will prevent the weeds and the movement of air in the soil will be good. Moisture will retain for a longer period in the soil. Weeding and digging will also help the growth of the roots. Germination is more.

8.2.5 Irrigation



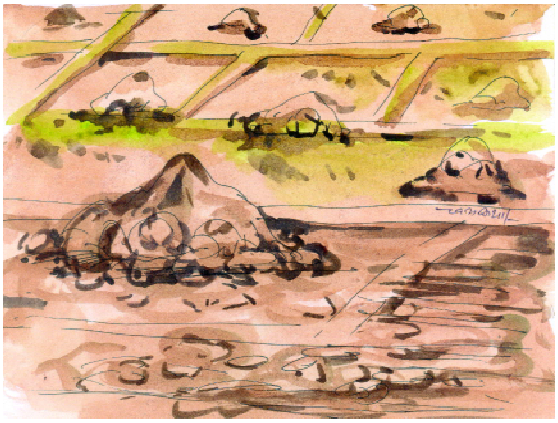
Irrigation

Every crop needs water, some more some less. We have to irrigate the crop from time to time as required by it. But remember, the time and quantity of water will depend on the crop. It will be harmful for the crop it the water given is move than required.

8.2.6 Compost and Fertilizer

All crops need proper amount of nutrients. These nutrients are given in the form of compost and fertilizers. There are many types of chemical fertilizers available in the market.

Using fertilizers thoughtlessly and excessively harms both the crop and the farmland. Therefore before sowing the crop, the soil of the farm should be sent for a check. This will tell us it constituents and its quantity present in the soil and which other constituent is needed by the crop which is to be sown in that soil. Use the fertilizers accordingly. Compost manure and other organic manures are beneficial for crops. Organic manure increases the yield and also the cereals, vegetables and fruits grown are tastier and healthier. Therefore, as far as possible, use organic manure. In bio-fertilizers, besides organic manure and cow dung. Organic manure consists of nadep compost, vermi compost, oil cake compost and green manure also.



Compost

Remember these sayings of Ghagh on the importance of organic manure.

1) Khad Padhe Toh Khet
   Nahi Toh Kooda Rhet
   (If there is manure, it is a field,
   Otherwise a garbage of sand.)

2) Gobbar, Maila, Neem Ki Khalli,
   Inse Kheti Doorni Phalli.
   (Cowdung, humus, manure of Neem leaves,
   Gives you a double yield.)

3) Joike Khet Padha Nahi Gobbar,
Wahi Kisan Ko Jano Doobar
(Anyone who has not put
cow dung manure in his field
That farmer will be thought of as weak.)

4) Gobbar, Maila, Paati Sade,
Tab Kheti Mei Dana Pade
(When there is cow dung, humus and decomposed leaves,
Then the grain in the field will appear)

5) Khete Pansa Je Na Kisan
Usko Kare Daridra Samman
(A farmer who does not put compost in his farm
will become a poor man)

8.2.7 Protection of Crops

It is generally noticed that if there is some kind an outbreak of disease or pests on the crops. They can cause massive damage to the harvest. Sometimes the entire crop is wasted. Thus from time to time we must identify the disease and the pests and take steps to protect the crops from them. For this a varity of chemicals, pesticides are available in the market, but if we use them blindly then it can be very harmful. Apart from chemicals there are a number organic treatments which if adopted will provide greater safety. They do not harm our health, the environment remains clean and money is also saved.



Use of Pesticides



Protection from

Apart from pests and diseases, neel gai, monkeys, birds and wild animals also destroy the crops. We should take appropriate steps to protect our crops and get a good yield otherwise all our efforts will be undone. Our lush green harvest will be ruined in no time.

8.2.8 Harvesting and Sorting

For any crop, harvesting and sorting is the final execution of work which brings to us the fruits of our labour. Harvesting should always be done at the right time. We should not harvest before the grain ripens they can be spoilt soon.

If we delay in harvesting then grains of the ripened crops can drop on the ground leading to loss of crop. After harvesting the crop, its sorting is also important. At the time of sorting the crop should be dry and crunchy. After sorting, the crop should be dried in the sun and then it is ready for storage.



Sorting

Let's see, what you have learnt 8.2

I. What happens if the crop is weak?

_____

II. When should we cultivate the land?

_____

III. From which things do the crops need protection?

_____

IV. What are the disadvantages of late harvesting?

_____

8.3 Weather and Crops

In our country, we sow all kinds of crops throughout the year. These crops have their different needs. All crops cannot be sown in all weathers and same climate. Therefore, it is important that we have information about it. The sowing of wrong crop at the wrong time will lead to losses only.

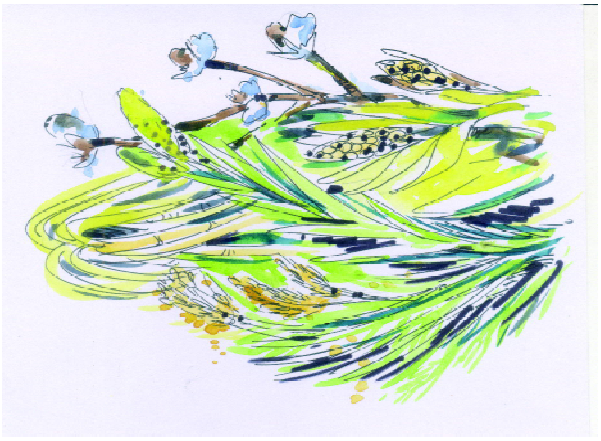
8.3.1 Which crop in which weather?

For every crop to sprout grow and ripen, a certain temperature is required. Some crops need cool weather while others flourish in the hot weather. In the same way, some crops need more water and humid weather.

Keeping this in mind we have divided the crops according to the weather in three parts.

8.3.2 Rabi Crop

The main crops of the Rabi season are wheat, oats, gram, peas, linseed, rye, mustard, potatoes etc. Their seeds sprout will in slightly cold weather. For the plants to expand they need more cold weather. For the crop to ripen, it needs to be slightly warm. These crops are therefore gown at the beginning of cold weather i.e. November - December. The crop is harvested when the hot season begins i.e. March - April.



Rabi Crops

8.3.3 Kharif Crops

Rice, maize, millets, moong, urad, sugarcane, cotton, tobacco etc are the kharif crops. These crops need lots of water, humidity and hot weather. Therefore they are sown as soon as the monsoon start i.e. July - August and the harvesting starts at the onset of cold weather i.e. October - November.

8.3.4 Zayyad Crops (Additional Crops)



After the harvesting of rabi crop and before the sowing of kharif crop the fields are generally vacant. From March - April till the end of June the empty farms are utilized by sowing some other crops. These crops are sunflower, cucumber, melon, water melon, gourd, pumpkin, lady finger, brinjals, ridge gourd etc. They are called zayyad crop.

They are sown in March - April and harvested before the rainy season starts.

Let's see, what you have learnt 8.3

I. When is the rabi crop sown?

_____

II. What is sown for karif crop?

_____

III. In which season are the seeds of water melon sown?

_____

Let's Revise

- In our country, four major types of soil are found - clay soil, sandy soil, alluvial soil and acidic soil.
- Humus makes the soil soft and fertile. The top layer of the soil is most fertile. It is called the wealth of the farmer.
- To grow any crop, some important work has to be done from the beginning to the end - preparing the farm, sowing, weeding and hoeing, irrigating the farm, putting manure and fertilizers, security of crop and cutting and sorting.
- There are three seasons for crops - Rabi, Kharif and Zayyad.

Exercise

I. Read and answer the questions.

There is a saying by Goswami Tulsidas

"Ka barsa Jab Krishi Sukhane"

"Samay Chuki Puni Ka Pachitane"

Means - what is use of rains if the crop has dried up. If you do something, if its time has slipped by, then it is of no use. We just repent. This is true of farming also. If we plant seeds after the sowing season then the seed will not germinate properly. If we don't do weeding and hoeing on time then weeds will stifle it. If we don't give water to the fields

when needed, the crops will dry up.

A good farmer is one who does his farming on time.

i. What will happen if you sow when the time has slipped by?

_____

ii. What will happen if we do not do weeding and hoeing on time?

_____

iii. What will happen if we do not irrigate the fields when they need water?

_____

iv. Who is a good farmer?

_____

II. Give the answer to the questions given below?

i. Which soil is most suitable for growing, potatoes, water-melon, sweet potatoes and groundnuts?

_____

ii. At the time of sorting how should the harvest crop look like?

_____

iii. In which season is gram cultivated?

_____

iv. When is zayyad crop sown?

_____

III. Which crop will be grown in which season? Choose the crops given below and write according to the season given.

Gram, Millet, Rice, Maize, Potato, Jowar, Cotton, sunflower, Water - Melon, Jute, Wheat, Cucumber, Tobacco, Linseed, Rye, Urad, Peas.

Rabi Crop _____

_____

Kharif Crop _____

_____

Zayyad Crop _____

_____

IV. Fill in the blanks by choosing the right word.

i. The top layer of soil is most ..... (Saleable / Fertile)

ii. In sandy soil it is easy to do weeding, hoeing and ..... (Cultivation / Whitewashing)

iii. The work of sowing must always be on ..... (time / home)

iv. The soil will dry up when there is ..... (Hurt / Oat)

v. Cow-dung, grime and cover of neem gives you double the ..... (Burnt / Fruit)

V. Match the following proverbs.

• Do not plough a raw farm	Gives you double yield.
• If there is manure then it is a field	Then the grains will appear.
• Cow-dung, humus and cover of leaves	That farmer will be considered weak
• Anyone who has not put cow-dung manure	It will yield neither seeds nor sprouts
• When there is cow-dung, humus and rotten leaves	Otherwise a waste of sand.

VI. Why is weeding and hoeing important? What are its benefits?

_____

_____

_____

_____

_____

_____

Come, let's do

- Make a small bed. Prepare it well for sowing.
- Make different lines on these beds. Sow 10 -10 seeds of rice, wheat, peas, maize, sunflower and water - melon.
- Observe how many plants of each seed grow.
- Take care of these plants - weeding, hoeing and irrigation.
- Look which plants are growing properly and which are dying.
- Why is it so? Consider it. Is it due to the season? Due to the soil? Because of hot - cold weather? Any other reason?

ANSWERS

Let's see, what you have learnt

8.1 I. Top layer of the soil.

II. Clay soil

III. Sandy soil

IV. Put lime and gypsum.

8.2 I. Decrease of yield.

II. Because of frost

III. Pests, diseases, neelgai, monkeys, birds and small animals.

IV. The seeds drop in the field.

8.3 I. November - December.

II. Rice, Maize, Jowar, Millets, Moong, Urad, Sugarcane, Cotton, Tobacco, Jute etc.

III. In zayyad

Exercise

I. i. Germination will be less.

ii. Weeds will crush the yield.

iii. The crop will dry up.

iv. One who does all the work of farming on time.

II. i. Sandy soil

ii. Dry and Crunchy

iii. of Rabi

iv. In March - April

III. Rabi Crop - Gram, Wheat, Peas, Rye, Linseed, Potato.

Kharif Crop - Millet, Rice, Maize, Jowar, Cotton, Jute, Tobacco, Urad.

Zayyad Crop - Sunflower, Water - Melon, Cucumber.

IV. i. Fertile

ii. Cultivating

iii. Time

iv. Frost (oat)

v. Grow

V.
- One should not plough a raw field. He will get neither seeds nor sprouts.
- It is a field, if man use is used. Otherwise it is garbage sand.
- Cow dung waste, cover of Neem. They lead to double yield.
- That farmer is a weak person, whose fields did not get cow dung.
- Grain is formed in crop, if cow dung, waste and rotten leaves are used as manure.

9

Conservation of Soil and Forests



We will learn in this lesson

- Soil erosion and what is soil erosion?
- How soil erodes?
- Harmful effects of soil erosion?
- How to prevent soil erosion?
- What is soil pollution? How can we stop it?
- Why protection of forests is necessary?

Soil is called a farmer's wealth. Any harm to the soil means loss to the crops and loss to a farmer. Overall it is a loss to the nation. That is why it is important that the soil is clean, healthy and in good condition. Like soil, the importance of forests is no less on the earth. Forests are our natural inheritance. Destruction of forest means our own ruin. Therefore, protection of forests and saving them from being destroyed are our duty. Come; let us know what we can do to protect our soil and forests?

9.1 Soil erosion and ways to prevent it

Do you know the top layer of the soil is most valuable for crops? Just think, if this top layer gets

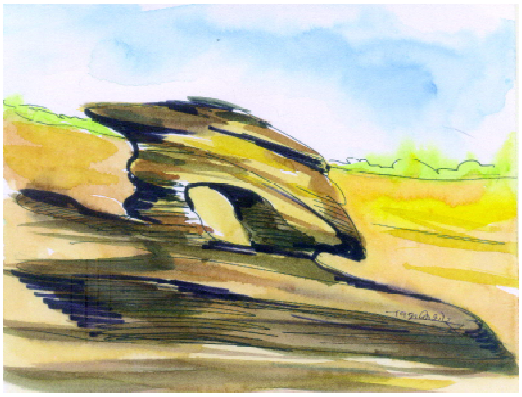
destroyed for some reason, then what will happen? Obviously the yield of the crop will decrease drastically. This layer of the soil does not get destroyed by itself. This is due to some natural reasons and also because of our carelessness that this soil erodes and moves from one place to another. The soil wearing away from its place is called degradation of soil or soil erosion. It is called de-layering also. There are two reasons for soil erosion - rain-water and wind.

When the rain water flows rapidly then the soil particles detach and flow along with it. At some places the soil erodes in the form of gullys or as flat surface. Gully erosion occurs when water rapidly flows in narrow channels and forms small gullys. At times, these gullys, after a long period of erosion, form big streams and ravines.



Erosion by water

Wind can also be a source of land degradation in many regions. Strong winds and gale pick up and carry away loose particles of soil. These particles also slowly erode any rock that comes in the way. Wind erosion is more common in arid and dry areas.



Erosion by wind

- Top layer of soil is most fertile. The loss of the top soil will adversely affect the growth and yield of crops.
- The wearing away of the field's top soil decreases its capacity to soak water. This reduces the water level in waterfalls, wells and in the earth.
- The eroded soil flows with water and accumulates at different places. This creates problems, for example, if it deposited in canals and sewers it obstructs the flow of water.
- When the soil is deposited in rivers this increases the danger of floods.
- There is loss of water in the rivers and smaller rivers may dry up.
- Erosion reduces grass in the pastures. Animals are not able to get hay to fill their belly to full capacity.
- When there is shortfall in farm yield the farmers suffer loss in their income.

How to prevent soil erosion

To stop all the losses arising from soil erosion, it is important to have effective methods to prevent it. These efforts and solutions are called soil conservation. Some methods of soil conservations are-

1. Making a boundary

It is called dolbandi also. On all the four sides of the fields a high wall is raised to prevent the fertile top soil and manure from flowing away.



Making a Boundary

2. Making an embankment

Making embankments on farms which are on slopes can prevent soil erosion. There should be a provision for an outlet to let the water flow out.



Embankment

3. Terrace farming

The water flows rapidly in the farms which are on steep slopes, and therefore soil erosion is quick. In such farms terrace farming is the practice. Terraces form a series of steps, each at a higher level than the previous one to stop erosion.



Terrace farming

4. Conservation of forests

Forests and Jungles are very effective in stopping soil erosion. Cutting and uprooting of trees increases soil erosion and destroys the shelter of flora and fauna. Therefore we must protect our forests. Sow grass, shrubs and trees in vacant land. The roots of the vegetation hold the soil and it is not swept away by water or wind - the two main reason of soil erosion.



Planting of Sapling

5. Making check-dam on rivers and drainages



Check Dams

Making check dams on rivers and sewers can reduce the flow of water and thereby reduce soil erosion. Check dams should have provision for water to flow out. The water will flow out gradually and the soil in it will deposit in the surrounding areas.

6. Make provisions for drainage of water

Water can not be retained in a farm for a very long time.
There is danger of the land becoming saline. There should be proper methods of drainage of water. Plant grass on the drains and boundaries of fields.

In this way, in different places and circumstances, soil erosion can be prevented by various methods. Remember the top soil of the field is very valuable and fertile for crops. This is the farmer legacy. We must protect this legacy.

Let's see, what you have learnt 9.1

I. What is land de-layering?

_____

II. What are the causes of soil erosion?

_____

III. How soil erosion can be prevented in places with steep slopes?

_____

IV. How are forests helpful in stopping soil erosion?

_____

9.2 Soil Pollution

Soil pollution means damage to the soil. Do you know how the soil is damaged?

When poisonous and harmful contents mix with the soil, then the soil gets polluted. What are these substances and who is responsible for this? Come, let's find out.

Causes of soil pollution

- The waste that comes from our homes, shops, lanes and roads also contains polythene, glass and iron pieces. When this waste is thrown here and there on the surface it mixes with the soil. It does not rot and decay. This harms the soil.

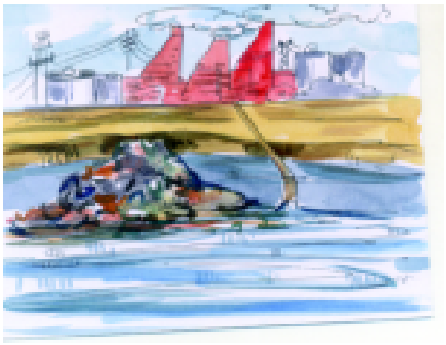


- To protect farms and orchards from harmful pests, chemical pesticides are recklessly used. These chemicals mix with the soil and damage it.



- Chemical fertilizers are used indiscriminately and are a reason of soil pollution.

- Some factories dispose of their waste in rivers and sewers. Their poisonous substances make the water of rivers and drainages also poisonous. This waste has a number of harmful and poisonous elements which reach the farms and makes the soil poisonous. These elements affect the yield of the fertile land.



Causes of Soil

If we look at it in this way, we all are responsible for soil pollution and the problem is increasing because of our ignorance and mistakes. Soil pollution is decreasing the strength of the soil.

If the soil is wasted away then the crop yield will be low. Trees and plants do not flourish properly. The greenery on earth will be affected. Reduction of trees and plants harms the environment. Our awareness can protect the soil from getting polluted.

How to prevent soil pollution

- Items like polythene, glass, iron etc. found in household or outside waste should be segregated. Do not throw them in the farms or holes dug for keeping manure. They can be recycled and made into new things.
- Owners of factories should not allow poisonous and harmful waste to flow into river and drains.



Disposal of waste

- Do not sprinkle chemical pesticides on crops or trees and plants, instead try organic treatment.
- To protect the crops from pests we should use medicines made from cow-dung, cow-urine, neem, tulsi, dhatura, tobacco, green chillies, garlic, hing and matka manure for sprinkling. Apart from protecting the farms from pests they also make cereals, vegetables and fruits healthy and tasty.
- Indiscriminate use of chemical fertilizers is lethal for the soil. Instead we must use organic manure as much as possible.



Organic Manure



Treatment of Seeds

- Organic manure like compost, cow-dung, manure, nadep, vermicompost, khali manure etc., if used, will be extremely beneficial. It improves the quality of soil, there is an increase of humus and the crop yield is better.
- Before sowing organic fertilizers like dricoderma, raijobium and P.S.B culture, can be used for the treatment of seeds. This protects the crop from pests and diseases and the yield increases.
- Matka manure made from cow-dung and cow - urine is very useful for crops.
- Use neem khali and rotten leaves for manure.
- If we sprinkle the urine of the cow on the crops then it protects them from pests and diseases. It works as a manure and also increases the yield.
- Green manure is a boon for fields. Before sowing the main crop, sow the seeds which make green manure and plough it into the field.



Green Manure

In this way if we keep changing the methods of farming then we can prevent soil pollution to a large extent. Organic farming is good for the health of soil, man, animals and birds. Therefore, we should stop the reckless use of chemicals and fertilizers in our farming and adopt organic methods. This is in our interest.